Chapitre 7 : La comparaison de deux groupes

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1 Avant de commencer

Ce chapitre reprends les codes du chapitre 7 sur ***la comparaison de deux groupes*** du livre **Langage R et Statistiques** : *Initiation à l'analyse de données* publié par les éditions ENI (2ème édition).

```
library(tidyverse)
```

```
── Attaching core tidyverse packages ──────────────────────── tidyverse 2.0.0 ──
✔ dplyr     1.2.1     ✔ readr     2.2.0
✔ forcats   1.0.1     ✔ stringr   1.6.0
✔ ggplot2   4.0.3     ✔ tibble    3.3.1
✔ lubridate 1.9.5     ✔ tidyr     1.3.2
✔ purrr     1.2.2
── Conflicts ────────────────────────────────────────── tidyverse_conflicts()
──
✖ dplyr::filter() masks stats::filter()
✖ dplyr::lag()    masks stats::lag()
ℹ Use the conflicted package (<http://conflicted.r-lib.org/>) to force all
conflicts to become errors
```

2 Comparaison de proportion

2.1 Test paramétrique

```
prop.test(x = c(42, 68), n = c(139,  175))
```

```

	2-sample test for equality of proportions with continuity correction

data:  c(42, 68) out of c(139, 175)
X-squared = 2.1762, df = 1, p-value = 0.1402
alternative hypothesis: two.sided
95 percent confidence interval:
 -0.19795128  0.02512497
sample estimates:
   prop 1    prop 2 
0.3021583 0.3885714 
```

```
prop.test(x = c(42, 68, 24), n = c(139, 175, 135))
```

```

    3-sample test for equality of proportions without continuity correction

data:  c(42, 68, 24) out of c(139, 175, 135)
X-squared = 16.187, df = 2, p-value = 0.0003056
alternative hypothesis: two.sided
sample estimates:
   prop 1    prop 2    prop 3 
0.3021583 0.3885714 0.1777778 
```

2.2 Test exact de Fisher

```
(genre_ratio_startup <-
  matrix(
    c(1, 2, 5, 7),
    nrow = 2,
    dimnames =
      list(
        c("Comite", "Salaries"),
        c("Femme", "Homme")
      )
  ))
```

```
         Femme Homme
Comite       1     5
Salaries     2     7
```

```
fisher.test(genre_ratio_startup)
```

```

    Fisher's Exact Test for Count Data

data:  genre_ratio_startup
p-value = 1
alternative hypothesis: true odds ratio is not equal to 1
95 percent confidence interval:
  0.009917539 17.649421539
sample estimates:
odds ratio 
  0.716429 
```

reprise de l'exemple de la page d'aide de `fisher.test()` : table Agresti (2002, p. 57) Job Satisfaction

```r
(Job <- matrix(
  c(1,2,1,0, 3,3,6,1, 10,10,14,9, 6,7,12,11),
  4,
  4,
  dimnames =
    list(
      salaire = c("< 15k", "15-25k", "25-40k", "> 40k"),
     satisfaction = c("tres decevant", "un peu decevant", "moderement satisfait",
"très satisfait")
    )
))
```

```
        satisfaction
salaire  tres decevant un peu decevant moderement satisfait très satisfait
  < 15k              1               3                   10              6
  15-25k             2               3                   10              7
  25-40k             1               6                   14             12
  > 40k              0               1                    9             11
```

```r
fisher.test(Job)
```

```


	Fisher's Exact Test for Count Data

data:  Job
p-value = 0.7827
alternative hypothesis: two.sided
```

2.3 Test de Mac Nemar

```r
(test_allergene <- matrix(
  c(23, 14, 84, 59),
  nrow = 2,
   dimnames = list(avant_traitement = c("réaction allergique", "réaction non
allergique"),
                       après_traitement =  c("réaction allergique", "réaction
non allergique")
  )
))
```

```
                          après_traitement
avant_traitement           réaction allergique réaction non allergique
  réaction allergique                       23                      84
  réaction non allergique                   14                      59
```

```r
mcnemar.test(test_allergene)
```

```

    McNemar's Chi-squared test with continuity correction

data:  test_allergene
McNemar's chi-squared = 48.582, df = 1, p-value = 3.168e-12
```

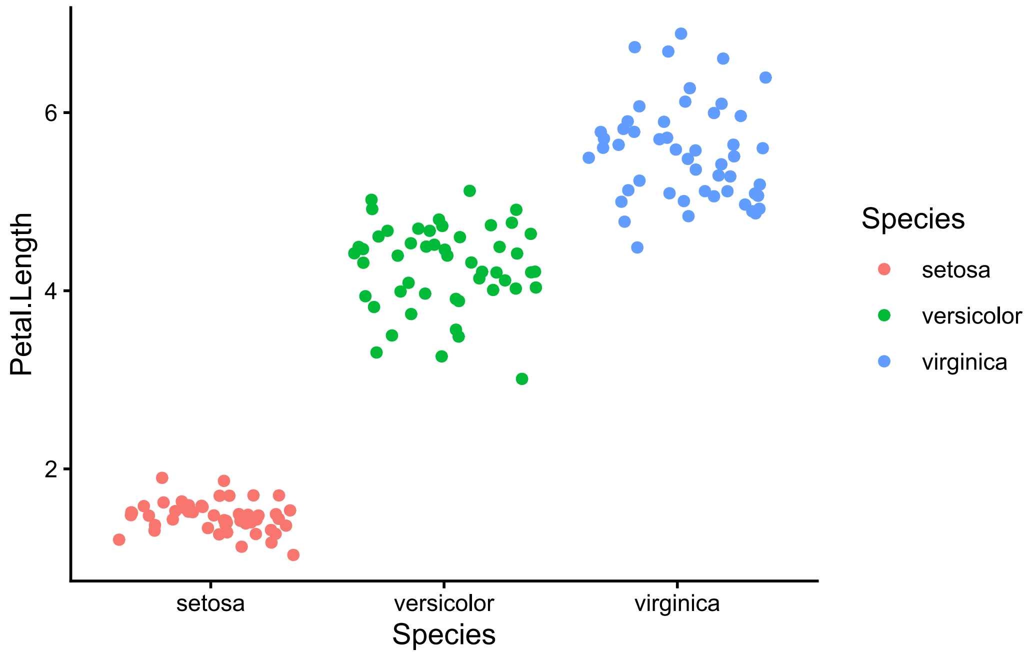
```r
prop.test(test_allergene)
```

```

    2-sample test for equality of proportions with continuity correction

data:  test_allergene
X-squared = 0.036067, df = 1, p-value = 0.8494
alternative hypothesis: two.sided
95 percent confidence interval:
 -0.1075760  0.1539209
sample estimates:
   prop 1    prop 2 
0.2149533 0.1917808 
```

3 Comparaison de variances

3.1 graphique initial

```r
ggplot(iris) +
  aes(x = Species, y = Petal.Length, color = Species) +
  geom_jitter() +
  theme_classic()
```



3.2 Test F

vérification de la normalité

```
filter(mtcars, am == 0)$mpg |> shapiro.test()
```

```

	Shapiro-Wilk normality test

data:  filter(mtcars, am == 0)$mpg
W = 0.97677, p-value = 0.8987
```

```
filter(mtcars, am == 1)$mpg |> shapiro.test()
```

```

	Shapiro-Wilk normality test

data:  filter(mtcars, am == 1)$mpg
W = 0.9458, p-value = 0.5363
```

test

```
var.test(filter(mtcars, am == 0)$mpg, filter(mtcars, am == 1)$mpg)
```

```

	F test to compare two variances

data:  filter(mtcars, am == 0)$mpg and filter(mtcars, am == 1)$mpg
F = 0.38656, num df = 18, denom df = 12, p-value = 0.06691
alternative hypothesis: true ratio of variances is not equal to 1
95 percent confidence interval:
 0.1243721 1.0703429
sample estimates:
ratio of variances 
         0.3865615 
```

3.3 Test de Bartlett

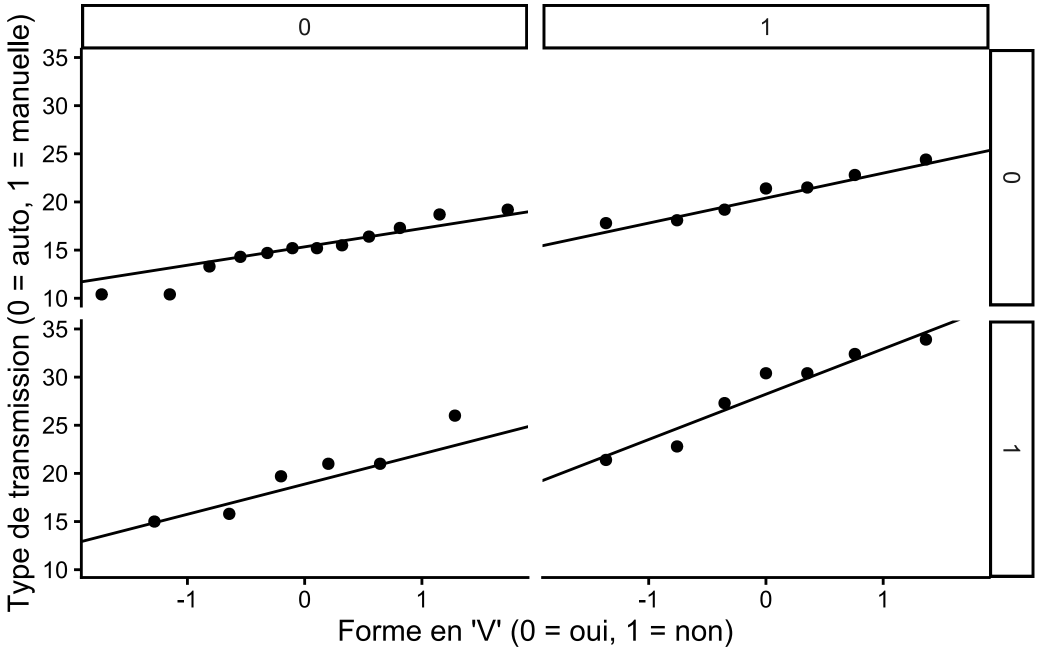
iris

```
bartlett.test(Petal.Length ~ Species, data = iris)
```

```

	Bartlett test of homogeneity of variances

data:  Petal.Length by Species
Bartlett's K-squared = 55.423, df = 2, p-value = 9.229e-13
```

mtcars : vérification de la normalité

```
ggplot(mtcars) +
  aes(sample = mpg) +
  geom_qq() +
  geom_qq_line() +
  facet_grid(rows = vars(am), cols = vars(vs)) +
   labs(x = "Forme en 'V' (0 = oui, 1 = non)", y = "Type de transmission (0 =
auto, 1 = manuelle)") +
  theme_classic()
```



mtcars : test de bartlett

```
bartlett.test(mpg ~ interaction(am, vs), data = mtcars)
```

```

	Bartlett test of homogeneity of variances

data:  mpg by interaction(am, vs)
Bartlett's K-squared = 3.5598, df = 3, p-value = 0.3131
```

3.4 Test de Levene

vérification de la normalité au sein de chaque espèce

```
filter(iris, Species == "setosa")$Petal.Length |> shapiro.test()
```

```

	Shapiro-Wilk normality test

data:  filter(iris, Species == "setosa")$Petal.Length
W = 0.95498, p-value = 0.05481
```

```
filter(iris, Species == "versicolor")$Petal.Length |> shapiro.test()
```

```

	Shapiro-Wilk normality test

data:  filter(iris, Species == "versicolor")$Petal.Length
W = 0.966, p-value = 0.1585
```

```r
filter(iris, Species == "virginica")$Petal.Length |> shapiro.test()
```

```

	Shapiro-Wilk normality test

data:  filter(iris, Species == "virginica")$Petal.Length
W = 0.96219, p-value = 0.1098
```

```r
library(car)
```

```
Le chargement a nécessité le package : carData
```

```


Attachement du package : 'car'
```

```
L'objet suivant est masqué depuis 'package:dplyr':

    recode
```

```
L'objet suivant est masqué depuis 'package:purrr':

    some
```

```r
leveneTest(Petal.Length ~ Species, data = iris)
```

```
Levene's Test for Homogeneity of Variance (center = median)
       Df F value    Pr(>F)    
group   2   19.48 3.129e-08 ***
      147                      
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

3.5 Test de Fligner-Killeen

vérification de la normalité des données pour chaque spray

```
purrr::map(
  .x = LETTERS[1:6],
  .f = ~ filter(InsectSprays, spray == .x)$count |> shapiro.test()
) |>
  set_names(glue::glue("Insecticide {LETTERS[1:6]}"))
```

```
$`Insecticide A`

    Shapiro-Wilk normality test

data:  filter(InsectSprays, spray == .x)$count
W = 0.95757, p-value = 0.7487


$`Insecticide B`

    Shapiro-Wilk normality test

data:  filter(InsectSprays, spray == .x)$count
W = 0.95031, p-value = 0.6415


$`Insecticide C`

    Shapiro-Wilk normality test

data:  filter(InsectSprays, spray == .x)$count
W = 0.85907, p-value = 0.04759


$`Insecticide D`

    Shapiro-Wilk normality test

data:  filter(InsectSprays, spray == .x)$count
W = 0.75063, p-value = 0.002713


$`Insecticide E`
```

```
    Shapiro-Wilk normality test

data:  filter(InsectSprays, spray == .x)$count
W = 0.92128, p-value = 0.2967


$`Insecticide F`

    Shapiro-Wilk normality test

data:  filter(InsectSprays, spray == .x)$count
W = 0.88475, p-value = 0.1009
```

test

```r
fligner.test(count ~ spray, data = InsectSprays)
```

```

    Fligner-Killeen test of homogeneity of variances

data:  count by spray
Fligner-Killeen:med chi-squared = 14.483, df = 5, p-value = 0.01282
```

4 Comparaison de moyennes

4.1 Test de Student

```r
t.test(filter(mtcars, am == 0)$mpg, filter(mtcars, am == 1)$mpg, var.equal =
TRUE)
```

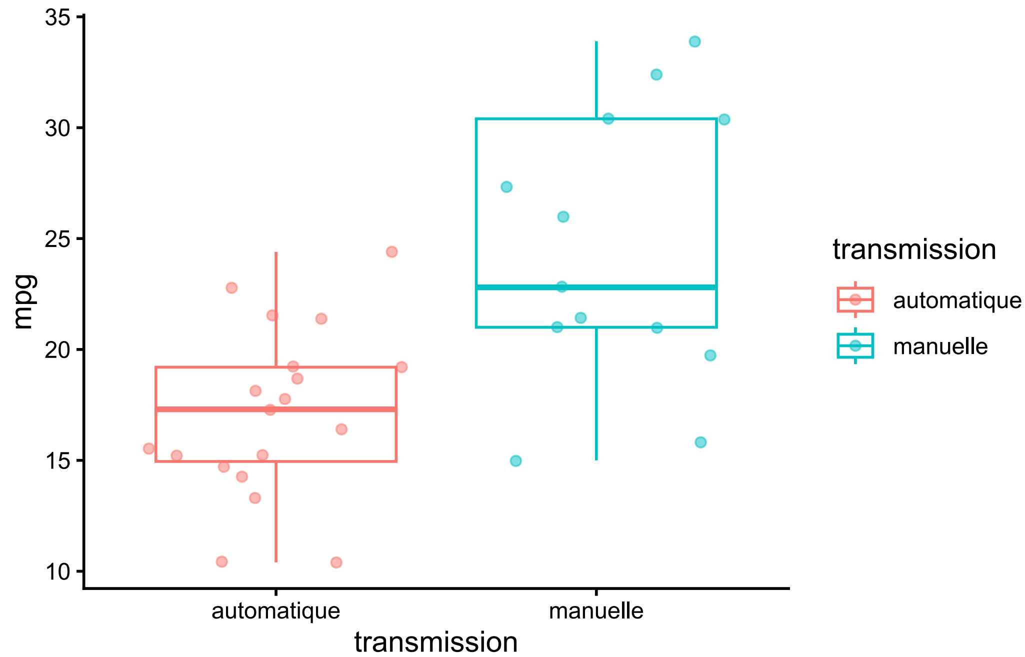
```

    Two Sample t-test

data:  filter(mtcars, am == 0)$mpg and filter(mtcars, am == 1)$mpg
t = -4.1061, df = 30, p-value = 0.000285
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 -10.84837  -3.64151
sample estimates:
mean of x mean of y
 17.14737  24.39231
```

```r
t.test(filter(mtcars, am == 0)$mpg, filter(mtcars, am == 1)$mpg, alternative =
"less", var.equal = TRUE)
```

```

	Two Sample t-test

data:  filter(mtcars, am == 0)$mpg and filter(mtcars, am == 1)$mpg
t = -4.1061, df = 30, p-value = 0.0001425
alternative hypothesis: true difference in means is less than 0
95 percent confidence interval:
      -Inf -4.250255
sample estimates:
mean of x mean of y
 17.14737  24.39231
```

graph

```r
mtcars |>
  mutate(
    transmission =
      case_when(
        am == 0 ~ "automatique",
        am == 1 ~ "manuelle"
      )
    ) |>
  ggplot() +
  aes(x = transmission, y = mpg, color = transmission) +
  geom_boxplot() +
  geom_jitter(alpha = 0.5) +
  theme_classic()
```



4.2 Test de Welch

```
t.test(Petal.Length ~ Species, data = iris |> filter(Species != "virginica"))
```

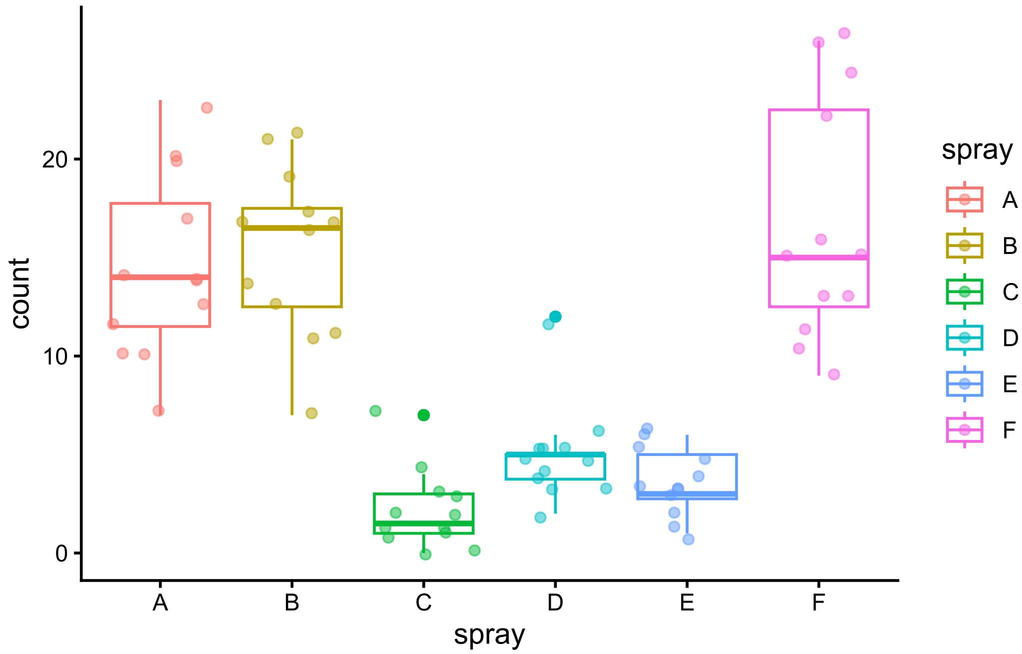
```


    Welch Two Sample t-test

data:  Petal.Length by Species
t = -39.493, df = 62.14, p-value < 2.2e-16
alternative hypothesis: true difference in means between group setosa and group
versicolor is not equal to 0
95 percent confidence interval:
 -2.939618 -2.656382
sample estimates:
    mean in group setosa mean in group versicolor
                   1.462                    4.260
```

4.3 Test de Wilcoxon

graph

```
InsectSprays |>
  ggplot() +
  aes(x = spray, y = count, color = spray) +
```

```
  geom_boxplot() +
  geom_jitter(alpha = 0.5) +
  theme_classic()
```



test de wilcoxon

```
wilcox.test(count ~ spray, data = InsectSprays |> filter(spray %in% c("E", "F")))
```

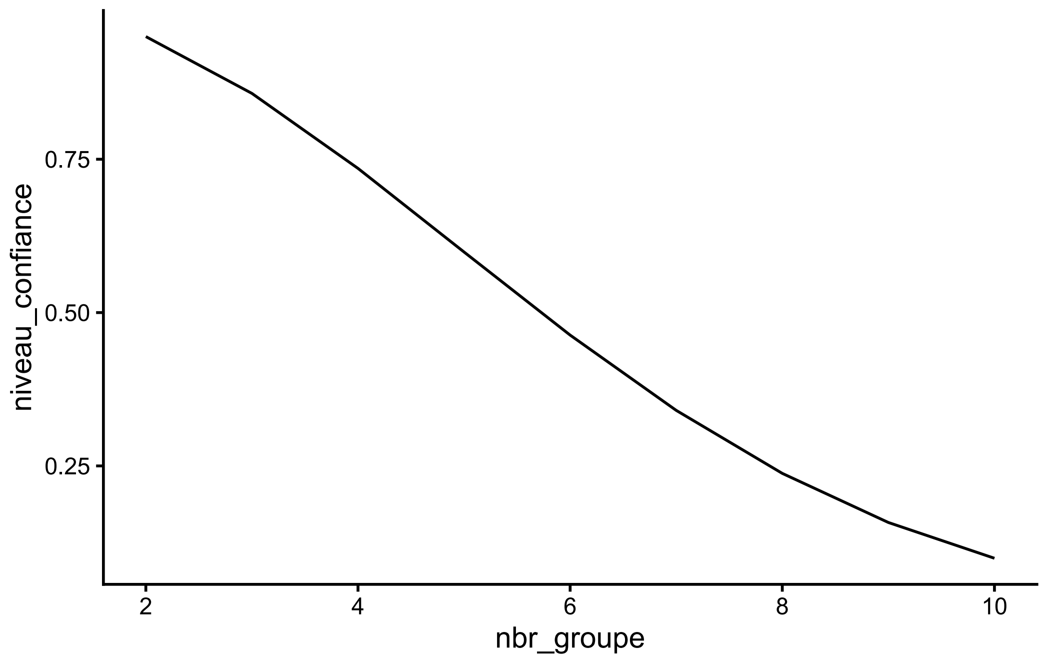
```


	Wilcoxon rank sum exact test

data:  count by spray
W = 0, p-value = 7.396e-07
alternative hypothesis: true location shift is not equal to 0
```

5 Comparaison de moyenne de plus de deux groupes

5.1 Erreurs quand le nombre de test est multiplié

```
tibble(
  nbr_groupe = 2:10,
  niveau_confiance = (0.95)^choose(nbr_groupe, 2)
) |>
```

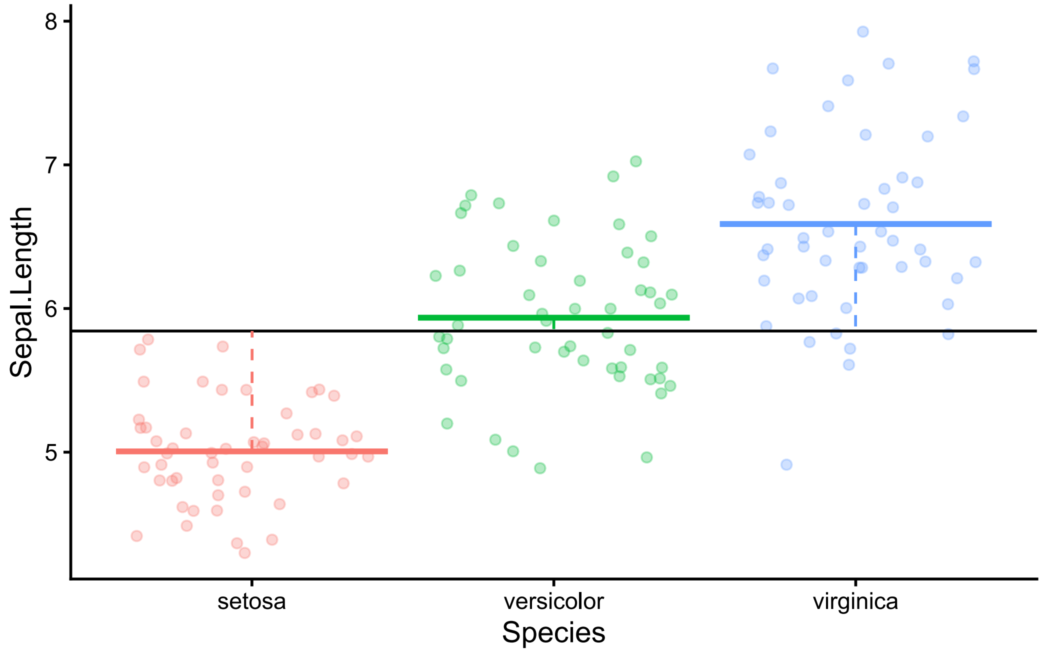
```
  ggplot() +
  aes(x = nbr_groupe, y = niveau_confiance) +
  geom_line() +
  theme_classic()
```



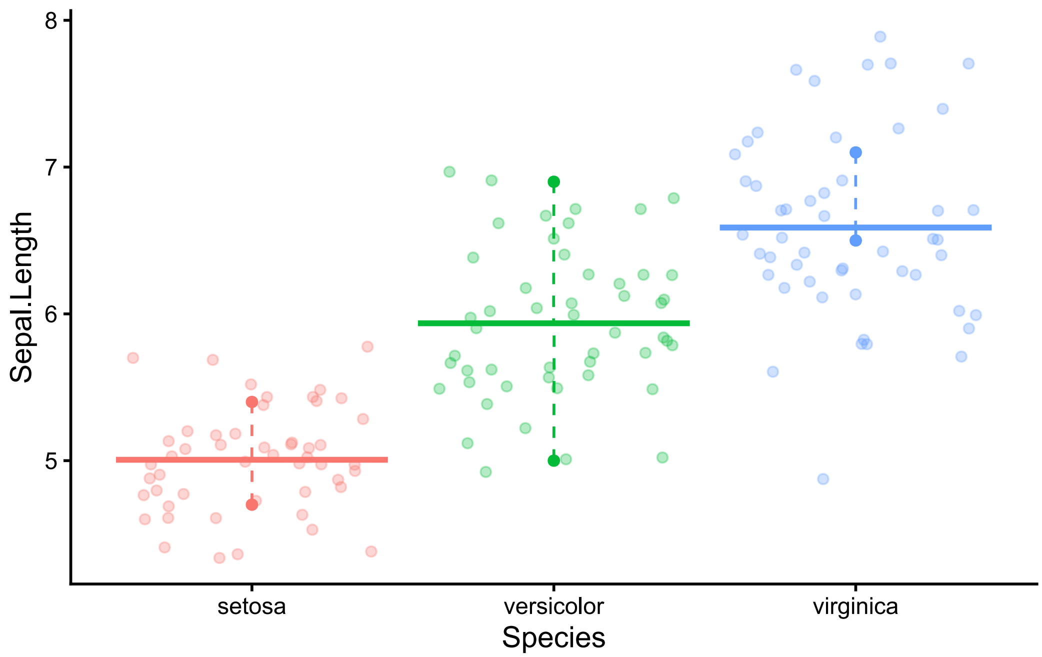
5.2 ANOVA

graph explication ANOVA

```
iris_moyenne <- iris |>
  group_by(Species) |>
  summarise(moyenne = mean(Sepal.Length))

ggplot(iris) +
  aes(x = Species, y = Sepal.Length, color = Species) +
  geom_jitter(alpha = 0.3) +
  geom_hline(aes(yintercept = mean(Sepal.Length))) +
    geom_errorbar(aes(ymin  =  moyenne,  y  =  5.84,  ymax  =  moyenne),  data  =
iris_moyenne, linewidth = 1) +
  geom_spoke(aes(y = moyenne, radius = mean(iris$Sepal.Length) - moyenne, angle
= 1.57), data = iris_moyenne, linetype = "dashed") +
  theme_classic() +
  theme(legend.position = "none")
```

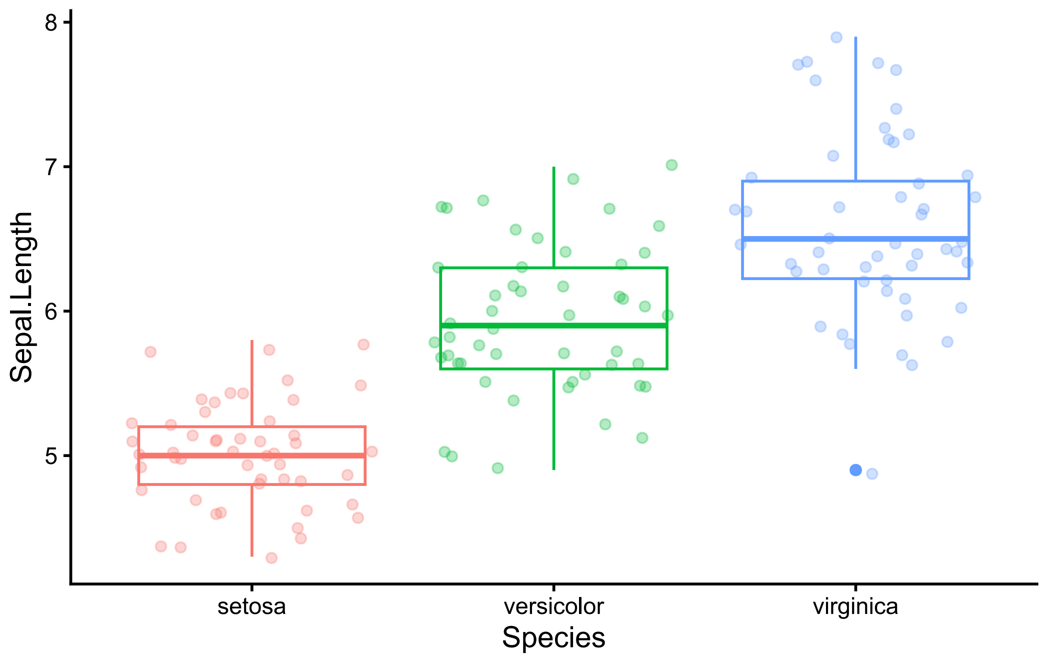



```r
ggplot(iris) +
  aes(x = Species, y = Sepal.Length, color = Species) +
  geom_jitter(alpha = 0.3) +
    geom_errorbar(aes(ymin  =  moyenne,  y  =  5.84,  ymax  =  moyenne),  data  =
iris_moyenne, linewidth = 1) +
   geom_spoke(aes(radius = iris_moyenne$moyenne - Sepal.Length, angle = 1.57),
data = iris |> group_by(Species) |> slice(3) |> ungroup(), linetype = "dashed") +
    geom_spoke(aes(radius = iris_moyenne$moyenne - Sepal.Length, angle = 1.57),
data = iris |> group_by(Species) |> slice(11) |> ungroup(), linetype = "dashed") +
  geom_point(data = iris |> group_by(Species) |> slice(c(3, 11)) |> ungroup()) +
  theme_classic() +
  theme(legend.position = "none")
```



vérification de la normalité et de l'homogénéité des variances

```
ggplot(iris) +
  aes(x = Species, y = Sepal.Length, color = Species) +
  geom_boxplot() +
  geom_jitter(alpha = 0.3) +
  theme_classic() +
  theme(legend.position = "none")
```



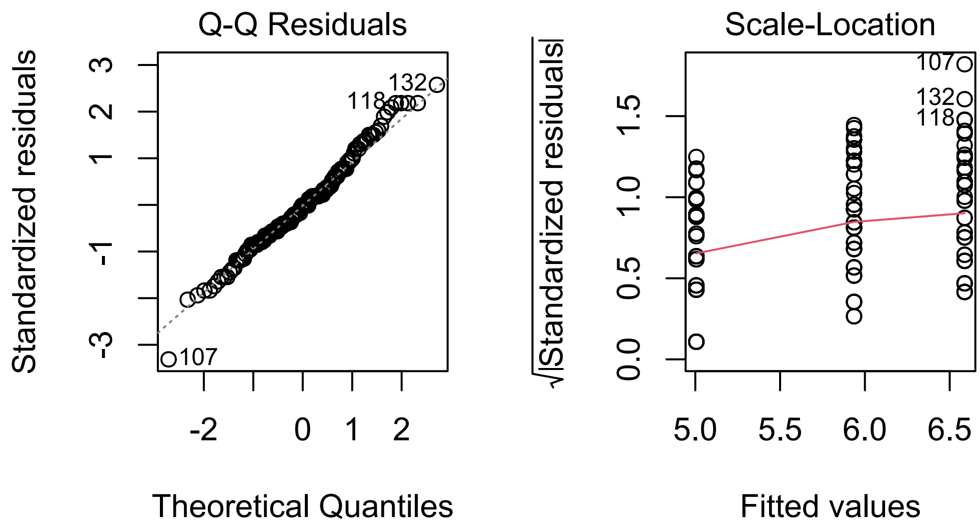
réalisation de l'ANOVA

```
iris_sepal_length_anova <- aov(Sepal.Length ~ Species, data = iris)
summary(iris_sepal_length_anova)
```

```
             Df Sum Sq Mean Sq F value Pr(>F)
Species       2  63.21  31.606   119.3 <2e-16 ***
Residuals   147  38.96   0.265
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

validation du test réalisé

```
par(mfrow=c(1,2))
plot(iris_sepal_length_anova, 2)
plot(iris_sepal_length_anova, 3)
```



post-hoc

```
rstatix::tukey_hsd(iris_sepal_length_anova)
```

```
# A tibble: 3 × 9
  term    group1     group2     null.value estimate conf.low conf.high    p.adj
* <chr>   <chr>      <chr>           <dbl>    <dbl>    <dbl>     <dbl>    <dbl>
1 Species setosa     versicolor          0    0.93     0.686     1.17  3.39e-14
2 Species setosa     virginica           0    1.58     1.34      1.83  3.00e-15
3 Species versicolor virginica           0    0.652    0.408     0.896 8.29e- 9
# ℹ 1 more variable: p.adj.signif <chr>
```

5.3 Test de Kruskal-Wallis

```
kruskal.test(count ~ spray, data = InsectSprays)
```

```


	Kruskal-Wallis rank sum test

data:  count by spray
Kruskal-Wallis chi-squared = 54.691, df = 5, p-value = 1.511e-10
```

Post-hoc de Nemenyi

```r
PMCMRplus::kwAllPairsNemenyiTest(count ~ spray, data = InsectSprays)
```

```
Warning in kwAllPairsNemenyiTest.default(c(10, 7, 20, 14, 14, 12, 10, 23, :
Ties are present, p-values are not corrected.
```

```


	Pairwise comparisons using Tukey-Kramer-Nemenyi all-pairs test with Tukey-
Dist approximation
```

```
data: count by spray
```

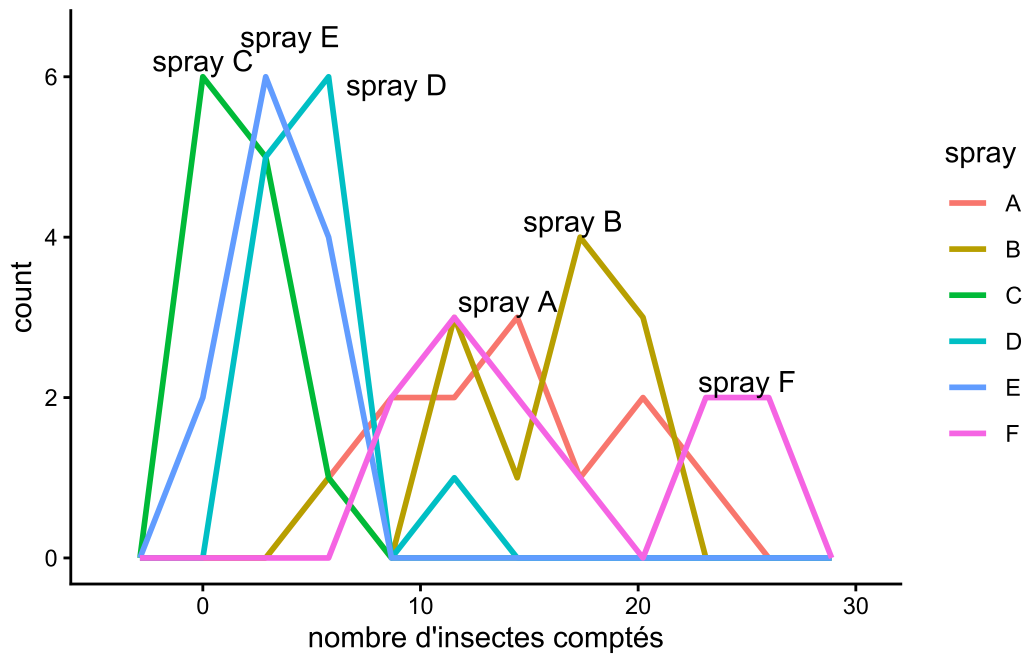
```
  A       B       C       D       E      
B 0.99961 -       -       -       -      
C 2.8e-05 5.7e-06 -       -       -      
D 0.02293 0.00813 0.56300 -       -      
E 0.00169 0.00047 0.94109 0.97809 -      
F 0.99861 1.00000 3.5e-06 0.00585 0.00031
```

```


P value adjustment method: single-step
```

```
alternative hypothesis: two.sided
```

graph

```r
ggplot(InsectSprays) +
  aes(color = spray, x = count) +
  geom_freqpoly(bins = 10, linewidth = 1) +
  labs(x = "nombre d'insectes comptés") +
  annotate("text", x = 0, y = 6.2, label = "spray C") +
  annotate("text", x = 4, y = 6.5, label = "spray E") +
  annotate("text", x = 8.9, y = 5.9, label = "spray D") +
  annotate("text", x = 17, y = 4.2, label = "spray B") +
  annotate("text", x = 14, y = 3.2, label = "spray A") +
  annotate("text", x = 25, y = 2.2, label = "spray F") +
  theme_classic()
```




```
ggsave("img/07Rl32N.png", width = 14, height = 10, units = "cm")
```